

WEBSERVICE DOCUMENTATION

INTERNATIONAL POSTCODE SYSTEM

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Description:

WebService has got six functions, which allow to get addresses from specify IPCS, find IPCS based on address, insert address to database, convert coordinates to and from IPCS, and calculate the distance between two IPCS codes.

WebService location:

<http://<domain>/WebService/>

Methods:

Get addresses from IPCS

<http://<domain>/WebService/indexAPI.php?action=getaddresses&ipcs=<IPCS>>

Parameters

Parameter	Type	Example	Description
action	String	getaddresses	Constant value.
ipcs	String	V9X0KFPC	Full or part of IPCS.

```
{
  "success":true,
  "addresses":[{
    "address":"64 Newbold Hall Drive",
    "town":"Rochdale",
    "country":"United Kingdom",
    "poi":"Hospital",
    "ipcs":"V9X0KFPC"
  }]
}
```

Find IPCS based on address

<http://<domain>/WebService/indexAPI.php?action=getpostcodes&address=<address>&town=<town>&ipcs=<postcode>&poi=<poi>&country=<count>>

Parameters

Parameter	Type	Example	Description
action	String	getpostcodes	Constant value.
address	String	Newbold	Full or part of address (optional).

town	String	Rochdale	Full or part of town (optional).
ipcs	String	V9X0	Full or part of ipcs (optional).
poi	String	Hosp	Full or part of point of interest (optional).
country	String	United	Full or part of country (optional).

```
{
  "success":true,
  "ipcs":["V9X0KFPC", "V9X0KFPB"]
}
```

Insert address. Address will be visible in the system after verification.

<http://<domain>/WebService/indexAPI.php?action=insertaddress&address=<address>&town=<town>&poi=<poi>&ipcs=<postcode>&country=<count>>

Parameters

Parameter	Type	Example	Description
action	String	insertaddress	Constant value.
address	String	Koszykowa 1	Full address.
town	String	Warszawa	Town or city.
poi	String	Hospital	Point of interest.
ipcs	String	1QYAVDSH	IPCS.
country	String	Poland	Country in english.

```
{
  "success":true
}
```

Generate IPCS from coordinates (geocode)

Converts a pair of geographic coordinates (longitude, latitude) into an 8-character International Postcode.

<http://<domain>/WebService/indexAPI.php?action=geocode&lon=<longitude>&lat=<latitude>>

Parameters

Parameter	Type	Example	Description
action	String	geocode	Constant value.
lon	Number	-2.0002	Longitude, decimal degrees (-180 to 180).
lat	Number	54.3222	Latitude, decimal degrees (-90 to 90).

Example:

<http://<domain>/WebService/indexAPI.php?action=geocode&lon=-2.0002&lat=54.3222>

```
{
  "success":true,
  "Postcode":"1GL8YT7F"
}
```

If the supplied coordinates are outside the valid range, or the request cannot be processed, the response is:

```
{
  "success":false,
  "error":"<error message>"
}
```

Get coordinates from IPCS (reverse geocode)

Converts a full 8-character International Postcode back into the coordinates of the centre of its sector.

<http://<domain>/WebService/indexAPI.php?action=reversegeocode&ipcs=<IPCS>>

Parameters

Parameter	Type	Example	Description
action	String	reversegeocode	Constant value.
ipcs	String	1GL8YT7F	Full 8-character IPCS.

Example:

<http://<domain>/WebService/indexAPI.php?action=reversegeocode&ipcs=1GL8YT7F>

```
{
  "success":true,
  "Lon":18.7715,
  "Lat":50.31025
}
```

If the IPCS is not exactly 8 characters long, or contains invalid characters, the response is:

```
{
  "success":false,
  "error":"Wrong postcode."
}
```

Calculate distance between two IPCS codes

Calculates the great-circle distance, in kilometres, between the centres of the sectors identified by two IPCS codes.

<http://<domain>/WebService/indexAPI.php?action=distance&from=<IPCS>&to=<IPCS>>

Parameters

Parameter	Type	Example	Description
action	String	distance	Constant value.
from	String	1GL8YT7F	Full 8-character IPCS of the starting point.

to	String	V9X0KFPC	Full 8-character IPCS of the destination.
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Example:

`http://<domain>/WebService/indexAPI.php?action=distance&from=1GL8YT7F&to=V9X0KFPC`

```
{
  "success":true,
  "Distance":245.187
}
```

If either IPCS code is invalid, the response is:

```
{
  "success":false,
  "error":"Distance cannot be calculated."
}
```

Navigation:

`http://<domain>?ipcs=<ipcs>`

`http://<domain>?coord=<latitude>,<longitude>`

`http://<domain>?lat=<latitude>&lng=<longitude>`

Examples:

`http://<domain>?ipcs=062WT853`

`http://<domain>?coord=53.344,0.22333`

`http://<domain>?lat=54.3222&lng=-2.0002`